JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Out of Syllabus Problems (2015–2021)

Category: Legacy & Discontinued Topics • Legacy Problem Collection

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
50	87 M	150 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	26	+1 / +2 / +6 Marks	−1/3 / −2/3 / −2 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	12	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	12	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Historical & Discontinued Topics:** This collection contains authentic official IIT JAM Mathematics questions from historical topics (such as Vector Calculus including Green's, Stokes', and Gauss Divergence Theorems, Mechanics, Statics, and Dynamics) that were tested in earlier years (2005–2021) but have been phased out of the modern JAM MA syllabus.
2. **Problem-Solving & Entrance Value:** These problems remain exceptionally valuable for competitive mathematical examinations (TIFR, GATE MA, CSIR-NET, NBHM, and CMI/ISI admissions) and deepening advanced analytical intuition.
3. **Section A (MCQ):** Four options (A), (B), (C), (D) where **only one** is correct. Incorrect attempts incur standard negative marking penalty.
4. **Section B (MSQ):** Four options (A), (B), (C), (D) where **one or more** options may be correct. Full marks awarded only if all correct choices and zero incorrect choices are chosen. No partial or negative marking.
5. **Section C (NAT):** Numerical answers are real numbers within the official accepted interval. No negative marking.
6. **Master Answer Key:** Complete official answers and acceptance bounds are provided on the final page(s) of this document.
7. **Online Interactive Simulator:** Practice full papers and topic test series at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q58 • ID: JAM_2021_Q58) NAT +2 M Neg: Nil (0)

Q. 58 Let S be the surface defined by

$$\{(x, y, z) \in \mathbb{R}^3 : z = 1 - x^2 - y^2,\ z \geq 0\}.$$

Let $\vec{F} = -y\hat{i} + (x-1)\hat{j} + z^2\hat{k}$ and $\hat{n}$ be the continuous unit normal field to the surface S with positive z-component. Then the value of

$$\frac{1}{\pi}\iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS$$

is _____.

Question 2 (JAM 2021 • Q49 • ID: JAM_2021_Q49) NAT +1 M Neg: Nil (0)

Q. 49 Let $\vec{F} = (y+1)e^y\cos(x)\hat{i} + (y+2)e^y\sin(x)\hat{j}$ be a vector field in $\mathbb{R}^2$ and C be a continuously differentiable path with the starting point $(0, 1)$ and the end point $(\frac{\pi}{2}, 0)$. Then

$$\int_C \vec{F} \cdot d\vec{r}$$

equals _____.

Question 3 (JAM 2021 • Q44 • ID: JAM_2021_Q44) NAT +1 M Neg: Nil (0)

Q. 44 Consider the subset $S = \{(x, y) : x^2 + y^2 > 0\}$ of $\mathbb{R}^2$. Let

$$P(x, y) = \frac{y}{x^2+y^2} \text{ and } Q(x, y) = -\frac{x}{x^2+y^2}$$

for $(x, y) \in S$. If C denotes the unit circle traversed in the counter-clockwise direction, then the value of

$$\frac{1}{\pi}\int_C (P\,dx + Q\,dy)$$

is _____.

Question 4 (JAM 2021 • Q37 • ID: JAM_2021_Q37) **MSQ** **+2 M** **Neg: Nil (0)**

Q. 37 Which of the following subsets of $\mathbb{R}$ is/are connected?

(A) The set $\{x \in \mathbb{R} : x \text{ is irrational}\}$.

(B) The set $\{x \in \mathbb{R} : x^3 - 1 \geq 0\}$.

(C) The set $\{x \in \mathbb{R} : x^3 + x + 1 \geq 0\}$.

(D) The set $\{x \in \mathbb{R} : x^3 - 2x + 1 \geq 0\}$.

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Question 5 (JAM 2021 • Q13 • ID: JAM_2021_Q13) **MCQ** **+2 M** **Neg: −0.67**

Q. 13 Consider the surface $S = \{(x, y, xy) \in \mathbb{R}^3 : x^2 + y^2 \leq 1\}$. Let $\vec{F} = y\hat{\imath} + x\hat{\jmath} + \hat{k}$. If $\hat{n}$ is the continuous unit normal field to the surface S with positive z-component, then

$$\iint_S \vec{F} \cdot \hat{n}\, dS$$

equals

(A) $\frac{\pi}{4}$.

(B) $\frac{\pi}{2}$.

(C) π.

(D) 2π.

Question 6 (JAM 2021 • Q8 • ID: JAM_2021_Q8) **MCQ** **+1 M** **Neg: −0.33**

Q. 8 Which one of the following subsets of $\mathbb{R}$ has a non-empty interior?

(A) The set of all irrational numbers in $\mathbb{R}$.

(B) The set $\{a \in \mathbb{R} : \sin(a) = 1\}$.

(C) The set $\{b \in \mathbb{R} : x^2 + bx + 1 = 0 \text{ has distinct roots}\}$.

(D) The set of all rational numbers in $\mathbb{R}$.

Question 7 (JAM 2020 • Q55 • ID: JAM_2020_Q55) **NAT** **+2 M** **Neg: Nil (0)**

Q. 55 Let C be the boundary of the square with vertices $(0,0)$, $(1,0)$, $(1,1)$ and $(0,1)$ oriented in the counter clockwise sense. Then, the value of the line integral

$$\oint_C x^2y^2dx + (x^2 - y^2)dy$$

is ________. (rounded off to two decimal places)

Question 8 (JAM 2020 • Q42 • ID: JAM_2020_Q42) **NAT** **+1 M** **Neg: Nil (0)**

Q. 42 Let $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ and S be the sphere given by $(x-2)^2 + (y-2)^2 + (z-2)^2 = 4$. If $\hat{n}$ is the unit outward normal to S, then

$$\frac{1}{\pi}\iint_S \vec{F}\cdot\hat{n}\, dS$$

is ________.

Question 9 (JAM 2020 • Q27 • ID: JAM_2020_Q27) **MCQ** **+2 M** **Neg: −0.67**

Q. 27 Let $D = \{(x,y) \in \mathbb{R}^2 : |x| + |y| \leq 1\}$ and $f : D \to \mathbb{R}$ be a non-constant continuous function. Which of the following is TRUE?

(A) The range of f is unbounded

(B) The range of f is a union of open intervals

(C) The range of f is a closed interval

(D) The range of f is a union of at least two disjoint closed intervals

Question 10 (JAM 2020 • Q26 • ID: JAM_2020_Q26) MCQ +2 M Neg: −0.67

JAM 2020 MATHEMATICS - MA

Q. 26 Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, x, y, z \in \mathbb{R}$. Which of the following is FALSE?

(A) $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$

(B) $\nabla \cdot (\vec{a} \times \vec{r}) = 0$

(C) $\nabla \times (\vec{a} \times \vec{r}) = \vec{a}$

(D) $\nabla \cdot ((\vec{a} \cdot \vec{r})\vec{r}) = 4(\vec{a} \cdot \vec{r})$

Question 11 (JAM 2019 • Q44 • ID: JAM_2019_Q44) NAT +1 M Neg: Nil (0)

Q.44 Let $\vec{F}(x, y) = -y\,\hat{\imath} + x\,\hat{\jmath}$ and let C be the ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

oriented counter clockwise. Then the value of $\oint_C \vec{F} \cdot d\vec{r}$ (round off to 2 decimal places) is_______________

Question 12 (JAM 2019 • Q35 • ID: JAM_2019_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Consider the intervals $S = (0, 2]$ and $T = [1, 3)$. Let S° and T° be the sets of interior points of S and T, respectively. Then the set of interior points of $S \setminus T$ is equal to

(A) $S \setminus T^\circ$

(B) $S \setminus T$

(C) $S^\circ \setminus T^\circ$

(D) $S^\circ \setminus T$

Question 13 (JAM 2019 • Q34 • ID: JAM_2019_Q34) MSQ +2 M Neg: Nil (0)

Q.34 Let $\vec{F}$ and $\vec{G}$ be differentiable vector fields and let g be a differentiable scalar function. Then

(A) $\nabla \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot \nabla \times \vec{F} - \vec{F} \cdot \nabla \times \vec{G}$

(B) $\nabla \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot \nabla \times \vec{F} + \vec{F} \cdot \nabla \times \vec{G}$

(C) $\nabla \cdot (g\vec{F}) = g\nabla \cdot \vec{F} - \nabla g \cdot \vec{F}$

(D) $\nabla \cdot (g\vec{F}) = g\nabla \cdot \vec{F} + \nabla g \cdot \vec{F}$

Question 14 (JAM 2019 • Q25 • ID: JAM_2019_Q25) MCQ +2 M Neg: −0.67

Q.25 Let $\vec{F}(x, y, z) = 2y\,\hat{\imath} + x^2\,\hat{\jmath} + xy\,\hat{k}$ and let C be the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 1$. Then the value of

$$\left| \oint_C \vec{F} \cdot d\vec{r} \right|$$

is

(A) π (B) $\frac{3\pi}{2}$ (C) 2π (D) 3π

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Question 15 (JAM 2019 • Q24 • ID: JAM_2019_Q24) MCQ +2 M Neg: −0.67

Q.24 Let C be the circle $(x-1)^2 + y^2 = 1$, oriented counter clockwise. Then the value of the line integral

$$\oint_C -\frac{4}{3}xy^3\,dx + x^4\,dy$$

is

(A) 6π (B) 8π (C) 12π (D) 14π

Question 16 (JAM 2019 • Q14 • ID: JAM_2019_Q14) MCQ +2 M Neg: −0.67

Q.14 The set $\left\{\frac{1}{m} + \frac{1}{n} :\ m, n \in \mathbb{N}\right\} \cup \{0\}$, as a subset of $\mathbb{R}$, is

(A) compact and open
(B) compact but not open
(C) not compact but open
(D) neither compact nor open

Question 17 (JAM 2019 • Q13 • ID: JAM_2019_Q13) MCQ +2 M Neg: −0.67

Q.13 The set $\left\{\frac{x}{1+x} :\ -1 < x < 1\right\}$, as a subset of $\mathbb{R}$, is

(A) connected and compact
(B) connected but not compact
(C) not connected but compact
(D) neither connected nor compact

Question 18 (JAM 2019 • Q5 • ID: JAM_2019_Q5) MCQ +1 M Neg: −0.33

Q.5 Let S be the set of all limit points of the set $\left\{\frac{n}{\sqrt{2}} + \frac{\sqrt{2}}{n} : n \in \mathbb{N}\right\}$. Let $\mathbb{Q}_+$ be the set of all positive rational numbers. Then

(A) $\mathbb{Q}_+ \subseteq S$
(B) $S \subseteq \mathbb{Q}_+$
(C) $S \cap (\mathbb{R} \setminus \mathbb{Q}_+) \neq \emptyset$
(D) $S \cap \mathbb{Q}_+ \neq \emptyset$

MA 3/13

Question 19 (JAM 2018 • Q40 • ID: JAM_2018_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Let S be a subset of $\mathbb{R}$ such that 2018 is an interior point of S. Which of the following is (are) TRUE?

(A) S contains an interval
(B) There is a sequence in S which does not converge to 2018
(C) There is an element $y \in S$, $y \neq 2018$ such that y is also an interior point of S
(D) There is a point $z \in S$, such that $|z - 2018| = 0.002018$

SECTION – C

NUMERICAL ANSWER TYPE (NAT)

Q. 41 – Q. 50 carry one mark each

Question 20 (JAM 2018 • Q39 • ID: JAM_2018_Q39) MSQ +2 M Neg: Nil (0)

Q.39 Which of the following subsets of $\mathbb{R}$ is (are) connected?

(A) $\{x \in \mathbb{R} \mid x^2 + x > 4\}$ (B) $\{x \in \mathbb{R} \mid x^2 + x < 4\}$

(C) $\{x \in \mathbb{R} \mid |x| < |x-4|\}$ (D) $\{x \in \mathbb{R} \mid |x| > |x-4|\}$

MA 9/12

Question 21 (JAM 2018 • Q38 • ID: JAM_2018_Q38) MSQ +2 M Neg: Nil (0)

Q.38 If $\vec{F}(x,y,z) = (2x+3yz)\hat{\imath} + (3xz+2y)\hat{\jmath} + (3xy+2z)\hat{k}$ for $(x,y,z) \in \mathbb{R}^3$, then which among the following is (are) TRUE?

(A) $\nabla \times \vec{F} = \vec{0}$

(B) $\oint_C \vec{F} \cdot d\vec{r} = 0$ along any simple closed curve C

(C) There exists a scalar function $\phi\colon \mathbb{R}^3 \to \mathbb{R}$ such that $\nabla \cdot \vec{F} = \phi_{xx} + \phi_{yy} + \phi_{zz}$

(D) $\nabla \cdot \vec{F} = 0$

Question 22 (JAM 2018 • Q34 • ID: JAM_2018_Q34) MSQ +2 M Neg: Nil (0)

Q.34 Let P and Q be two non-empty disjoint subsets of $\mathbb{R}$. Which of the following is (are) FALSE?

(A) If P and Q are compact, then $P \cup Q$ is also compact

(B) If P and Q are not connected, then $P \cup Q$ is also not connected

(C) If $P \cup Q$ and P are closed, then Q is closed

(D) If $P \cup Q$ and P are open, then Q is open

MA 8/12

Question 23 (JAM 2018 • Q20 • ID: JAM_2018_Q20) MCQ +2 M Neg: −0.67

Q.20 If $\vec{F}(x,y) = (3x - 8y)\hat{\imath} + (4y - 6xy)\hat{\jmath}$ for $(x,y) \in \mathbb{R}^2$, then $\oint_C \vec{F} \cdot d\,\vec{r}$, where C is the boundary of the triangular region bounded by the lines $x = 0, y = 0$ and $x + y = 1$ oriented in the anti-clockwise direction, is

(A) $\frac{5}{2}$ (B) 3 (C) 4 (D) 5

Question 24 (JAM 2018 • Q7 • ID: JAM_2018_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $f: \mathbb{R}^3 \to \mathbb{R}$ be a scalar field, $\vec{v}: \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field and let $\vec{a} \in \mathbb{R}^3$ be a constant vector. If $\vec{r}$ represents the position vector $x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, then which one of the following is FALSE?

(A) $curl(f\,\vec{v}) = grad(f) \times \vec{v} + f\;curl(\vec{v})$

(B) $div(grad(f)) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) f$

(C) $curl(\vec{a} \times \vec{r}) = 2\,|\vec{a}|\,\vec{r}$

(D) $div\left(\frac{\vec{r}}{|\vec{r}|^3}\right) = 0$, for $\vec{r} \neq \vec{0}$

Question 25 (JAM 2017 • Q52 • ID: JAM_2017_Q52) NAT +2 M Neg: Nil (0)

Q.52 Let T be the smallest positive real number such that the tangent to the helix

$$\cos t\;\hat{\imath} + \sin t\,\hat{\jmath} + \frac{t}{\sqrt{2}}\,\hat{k}$$

at $t = T$ is orthogonal to the tangent at $t = 0$. Then the line integral of $\vec{F} = x\hat{\jmath} - y\,\hat{\imath}$ along the section of the helix from $t = 0$ to $t = T$ is ____________

Question 26 (JAM 2017 • Q37 • ID: JAM_2017_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let S be the set of all rational numbers in $(0,1)$. Then which of the following statements is / are TRUE?

(A) S is a closed subset of $\mathbb{R}$

(B) S is not a closed subset of $\mathbb{R}$

(C) S is an open subset of $\mathbb{R}$

(D) Every $x \in (0,1)\setminus S$ is a limit point of S

Question 27 (JAM 2017 • Q24 • ID: JAM_2017_Q24) MCQ +2 M Neg: −0.67

Q.24 Let S be an infinite subset of $\mathbb{R}$ such that $S\setminus\{\alpha\}$ is compact for some $\alpha \in S$. Then which one of the following is TRUE?

(A) S is a connected set

(B) S contains no limit points

(C) S is a union of open intervals

(D) Every sequence in S has a subsequence converging to an element in S

Question 28 (JAM 2017 • Q18 • ID: JAM_2017_Q18) MCQ +2 M Neg: −0.67

Q.18 The flux of $\vec{F} = y\,\hat{\imath} - x\hat{\jmath} + z^2\hat{k}$ along the outward normal, across the surface of the solid

$$\left\{ (x,y,z) \in \mathbb{R}^3 \,\middle|\, 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le \sqrt{2 - x^2 - y^2} \right\}$$

is equal to

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$ (C) $\frac{8}{3}$ (D) $\frac{4}{3}$

Question 29 (JAM 2017 • Q17 • ID: JAM_2017_Q17) MCQ +2 M Neg: −0.67

Q.17 For $a > 0, b > 0$, let $\vec{F} = \frac{x\hat{j} - y\,\hat{\imath}}{b^2x^2+a^2y^2}$ be a planar vector field. Let

$$C = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = a^2 + b^2\}$$

be the circle oriented anti-clockwise. Then

$$\oint_C \vec{F} \cdot d\vec{r} =$$

(A) $\frac{2\pi}{ab}$ (B) 2π (C) $2\pi ab$ (D) 0

Question 30 (JAM 2017 • Q15 • ID: JAM_2017_Q15) MCQ +2 M Neg: −0.67

Q.15 The line integral of the vector field

$$\vec{F} = zx\,\hat{\imath} + xy\,\hat{\jmath} + yz\,\hat{k}$$

along the boundary of the triangle with vertices (1,0,0), (0,1,0) and (0,0,1), oriented anti-clockwise, when viewed from the point (2,2,2), is

(A) $\frac{-1}{2}$ (B) -2 (C) $\frac{1}{2}$ (D) 2

Question 31 (JAM 2017 • Q14 • ID: JAM_2017_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $\vec{F} = (3 + 2xy)\hat{\imath} + (x^2 - 3y^2)\hat{\jmath}$ and let L be the curve

$$\vec{r}(t) = e^t \sin t\,\hat{\imath} + e^t \cos t\,\hat{\jmath}, \qquad 0 \le t \le \pi.$$

Then

$$\int_L \vec{F} \cdot d\vec{r} =$$

(A) $e^{-3\pi} + 1$ (B) $e^{-6\pi} + 2$ (C) $e^{6\pi} + 2$ (D) $e^{3\pi} + 1$

Question 32 (JAM 2017 • Q11 • ID: JAM_2017_Q11) MCQ +2 M Neg: −0.67

Q.11 The flux of the vector field

$$\vec{F} = \left(2\pi x + \frac{2x^2y^2}{\pi}\right)\hat{\imath} + \left(2\pi xy - \frac{4y}{\pi}\right)\hat{\jmath}$$

along the outward normal, across the ellipse $x^2 + 16y^2 = 4$ is equal to

(A) $4\pi^2 - 2$ (B) $2\pi^2 - 4$ (C) $\pi^2 - 2$ (D) 2π

Question 33 (JAM 2016 • Q55 • ID: JAM_2016_Q55) NAT +2 M Neg: Nil (0)

Q.55 Let S be the closed surface forming the boundary of the region V bounded by $x^2 + y^2 = 3$, $z = 0$, $z = 6$. A vector field $\vec{F}$ is defined over V with $\nabla \cdot \vec{F} = 2y + z + 1$. Then the value of

$$\frac{1}{\pi}\iint_S \vec{F} \cdot \hat{n}\, ds,$$

where $\hat{n}$ is the unit outward drawn normal to the surface S, is ___________,

Question 34 (JAM 2016 • Q54 • ID: JAM_2016_Q54) NAT +2 M Neg: Nil (0)

Q.54 Let C be the boundary of the region enclosed by $y = x^2$, $y = x + 2$, and $x = 0$. Then the value of the line integral

$$\oint_C (xy - y^2)dx - x^3 dy,$$

where C is traversed in the counter clockwise direction, is ____________

MA 12/13

Question 35 (JAM 2016 • Q45 • ID: JAM_2016_Q45) NAT +1 M Neg: Nil (0)

Q.45 Let $\vec{F} = \sqrt{x}\,\hat{\imath} + (x+y^3)\hat{\jmath}$ be a vector field for all (x,y) with $x \geq 0$ and $\vec{r} = x\hat{\imath} + y\hat{\jmath}$. Then the value of the line integral $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(1,1)$ along the path $C\colon\ x = t^2,\ y = t^3,\ 0 \leq t \leq 1$ is ___________

Question 36 (JAM 2016 • Q40 • ID: JAM_2016_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Let $S = \left\{\frac{1}{3^n} + \frac{1}{7^m} \;\middle|\; n, m \in \mathbb{N}\right\}$. Then which of the following statement(s) is(are) TRUE?

(A) S is closed
(B) S is not open
(C) S is connected
(D) 0 is a limit point of S

SECTION – C

NUMERICAL ANSWER TYPE (NAT)

Q. 41 – Q. 50 carry one mark each.

Question 37 (JAM 2016 • Q36 • ID: JAM_2016_Q36) MSQ +2 M Neg: Nil (0)

Q.36 Which of the following statement(s) is(are) TRUE?

(A) There exists a connected set in $\mathbb{R}$ which is not compact
(B) Arbitrary union of closed intervals in $\mathbb{R}$ need not be compact
(C) Arbitrary union of closed intervals in $\mathbb{R}$ is always closed
(D) Every bounded infinite subset V of $\mathbb{R}$ has a limit point in V itself

Question 38 (JAM 2016 • Q34 • ID: JAM_2016_Q34) MSQ +2 M Neg: Nil (0)

Q.34 Consider the vector field $\vec{F} = x\hat{\imath} + y\hat{\jmath}$ on an open connected set $S \subset \mathbb{R}^2$. Then which of the following statement(s) is(are) TRUE ?

(A) Divergence of $\vec{F}$ is zero on S
(B) The line integral of $\vec{F}$ is independent of path in S
(C) $\vec{F}$ can be expressed as a gradient of a scalar function on S
(D) The line integral of $\vec{F}$ is zero around any piecewise smooth closed path in S

Question 39 (JAM 2016 • Q23 • ID: JAM_2016_Q23) MCQ +2 M Neg: −0.67

Q.23 Let $S \subset \mathbb{R}$ and ∂S denote the set of points x in $\mathbb{R}$ such that every neighbourhood of x contains some points of S as well as some points of complement of S. Further, let $\bar{S}$ denote the closure of S. Then which one of the following is FALSE?

(A) $\partial\mathbb{Q} = \mathbb{R}$
(B) $\partial(\mathbb{R} \setminus T) = \partial T, \quad T \subset \mathbb{R}$
(C) $\partial(T \cup V) = \partial T \cup \partial V, \quad T, V \subset \mathbb{R},\ T \cap V \neq \phi$
(D) $\partial T = \overline{T} \cap \overline{(\mathbb{R} \setminus T)}, \quad T \subset \mathbb{R}$

Question 40 (JAM 2016 • Q20 • ID: JAM_2016_Q20) MCQ +2 M Neg: −0.67

Q.20 Let S be the surface of the cone $z = \sqrt{x^2 + y^2}$ bounded by the planes $z = 0$ and $z = 3$. Further, let C be the closed curve forming the boundary of the surface S. A vector field $\vec{F}$ is such that $\nabla \times \vec{F} = -x\hat{\imath} - y\hat{\jmath}$. The absolute value of the line integral $\oint_c \vec{F} \cdot d\vec{r}$, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $r = |\vec{r}|$, is

(A) 0 (B) 9π (C) 15π (D) 18π

Question 41 (JAM 2016 • Q19 • ID: JAM_2016_Q19) MCQ +2 M Neg: −0.67

Q.19 Consider the vector field $\vec{F} = r^{\beta}(y\,\hat{\imath} - x\hat{\jmath}\,)$, where $\beta \in \mathbb{R}$, $\vec{r} = x\hat{\imath} + y\hat{\jmath}$ and $r = |\vec{r}|$. If the absolute value of the line integral $\oint_c \vec{F} \cdot d\vec{r}$ along the closed curve $C: x^2 + y^2 = a^2$ (oriented counter clockwise) is 2π, then β is

(A) -2 (B) -1 (C) 1 (D) 2

Question 42 (JAM 2016 • Q10 • ID: JAM_2016_Q10) MCQ +1 M Neg: −0.33

Q.10 Let S be a closed subset of $\mathbb{R}$, T a compact subset of $\mathbb{R}$ such that $S \cap T \neq \phi$. Then $S \cap T$ is

(A) closed but not compact
(B) not closed
(C) compact
(D) neither closed nor compact

Q. 11 – Q. 30 carry two marks each.

Question 43 (JAM 2016 • Q7 • ID: JAM_2016_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $\vec{r} = (x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k})$ and $r = |\vec{r}|$. If $f(r) = \ln r$ and $g(r) = \frac{1}{r}$, $r \neq 0$, satisfy $2\,\nabla f + h(r)\nabla g = \vec{0}$, then $h(r)$ is

(A) r (B) $\frac{1}{r}$ (C) $2r$ (D) $\frac{2}{r}$

Question 44 (JAM 2015 • Q51 • ID: JAM_2015_Q51) NAT +2 M Neg: Nil (0)

Q. 11 – Q. 20 carry two marks each.

Q.11 Let R be the planar region bounded by the lines $x = 0$, $y = 0$ and the curve $x^2 + y^2 = 4$, in the first quadrant. Let C be the boundary of R, oriented counter-clockwise. Then the value of

$$\oint_C x(1-y)\,dx + (x^2 - y^2)\,dy$$

is ______________

Question 45 (JAM 2015 • Q41 • ID: JAM_2015_Q41) NAT +1 M Neg: Nil (0)

Q. 1 – Q. 10 carry one mark each.

Q.1

Let C be the straight line segment from $P(0,\ \pi)$ to $Q\left(4,\ \frac{\pi}{2}\right)$, in the xy-plane. Then the value of $\int_C e^x(\cos y\,dx - \sin y\,dy)$ is ______________

Question 46 (JAM 2015 • Q34 • ID: JAM_2015_Q34) MSQ +2 M Neg: Nil (0)

Q.4

Let $\vec{F}$ be a vector field given by $\vec{F}(x,y,z) = -y\,\hat{\imath} + 2\,x\,y\,\hat{\jmath} + z^3\hat{k}$, for $(x,y,z) \in \mathbb{R}^3$. If C is the curve of intersection of the surfaces $x^2 + y^2 = 1$ and $y + z = 2$, then which of the following is (are) equal to $\left|\int_C \vec{F}\,.\,d\vec{r}\right|$?

(A) $\int_0^{2\pi}\int_0^1 (1 + 2r\sin\theta) r\,dr d\theta$

(B) $\int_0^{2\pi}\left(\frac{1}{2} + \frac{2}{3}\sin\theta\right)\,d\theta$

(C) $\int_0^{2\pi}\int_0^1 (1 + 2r\sin\theta)\ dr d\theta$

(D) $\int_0^{2\pi}(1 + \sin\theta)\,d\theta$

Question 47 (JAM 2015 • Q29 • ID: JAM_2015_Q29) **MCQ** **+2 M** **Neg: –0.67**

Q.29

Let G and H be nonempty subsets of $\mathbb{R}$, where G is connected and $G \cup H$ is not connected. Which one of the following statements is true for all such G and H ?

(A) If $G \cap H = \emptyset$, then H is connected

(B) If $G \cap H = \emptyset$, then H is not connected

(C) If $G \cap H \neq \emptyset$, then H is connected

(D) If $G \cap H \neq \emptyset$, then H is not connected

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Question 48 (JAM 2015 • Q20 • ID: JAM_2015_Q20) **MCQ** **+2 M** **Neg: –0.67**

Q.20

Let $S = \cap_{n=1}^{\infty}\left(\left[0, \frac{1}{2n+1}\right] \cup \left[\frac{1}{2n}, 1\right]\right)$. Which one of the following statements is **FALSE**?

(A) There exist sequences $\{a_n\}$ and $\{b_n\}$ in $[0,1]$ such that $S = [0,1] \setminus \cup_{n=1}^{\infty}(a_n, b_n)$
(B) $[0,1] \setminus S$ is an open set
(C) If A is an infinite subset of S, then A has a limit point
(D) There exists an infinite subset of S having no limit points

Question 49 (JAM 2015 • Q10 • ID: JAM_2015_Q10) **MCQ** **+1 M** **Neg: –0.33**

Q.10 Let A be a nonempty subset of $\mathbb{R}$. Let $I(A)$ denote the set of interior points of A. Then $I(A)$ can be

(A) empty
(B) singleton
(C) a finite set containing more than one element
(D) countable but not finite

Q. 11 – Q. 30 carry two marks each.

Question 50 (JAM 2015 • Q4 • ID: JAM_2015_Q4) MCQ +1 M Neg: −0.33

Q.4 Let S be a nonempty subset of $\mathbb{R}$. If S is a finite union of disjoint bounded intervals, then which one of the following is true?

(A) If S is not compact, then $\sup S \notin S$ and $\inf S \notin S$
(B) Even if $\sup S \in S$ and $\inf S \in S$, S need not be compact
(C) If $\sup S \in S$ and $\inf S \in S$, then S is compact
(D) Even if S is compact, it is not necessary that $\sup S \in S$ and $\inf S \in S$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Out of Syllabus Problems (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**2**
2	NAT	+1	**1**
3	NAT	+1	**−2**
4	MSQ	+2	**B, C**
5	MCQ	+2	**B**
6	MCQ	+1	**C**
7	NAT	+2	**0.65 to 0.67**
8	NAT	+1	**32 to 32**
9	MCQ	+2	**C**
10	MCQ	+2	**C**
11	NAT	+1	**75.35 to 75.45**
12	MSQ	+2	**B, D**
13	MSQ	+2	**A, D**
14	MCQ	+2	**C**
15	MCQ	+2	**B**
16	MCQ	+2	**B**
17	MCQ	+2	**B**

Q. No.	Type	Marks	Official Key
18	MCQ	+1	**B**
19	MSQ	+2	**A;B;C**
20	MSQ	+2	**B;C;D**
21	MSQ	+2	**A;B;C**
22	MSQ	+2	**B;C;D**
23	MCQ	+2	**B**
24	MCQ	+1	**C**
25	NAT	+2	**2.0 to 2.2**
26	MSQ	+2	**B, D**
27	MCQ	+2	**D**
28	MCQ	+2	**D**
29	MCQ	+2	**A**
30	MCQ	+2	**C**
31	MCQ	+2	**D**
32	MCQ	+2	**B**
33	NAT	+2	**72.0 to 72.0**
34	NAT	+2	**0.8 to 1.9**

Q. No.	Type	Marks	Official Key
35	NAT	+1	**1.49 to 1.55**
36	MSQ	+2	**B;D**
37	MSQ	+2	**A;B**
38	MSQ	+2	**B;C;D**
39	MCQ	+2	**C**
40	MCQ	+2	**MTA**
41	MCQ	+2	**A**
42	MCQ	+1	**C**
43	MCQ	+1	**C**
44	NAT	+2	**8**
45	NAT	+1	**1**
46	MSQ	+2	**A;B**
47	MCQ	+2	**D**
48	MCQ	+2	**D**
49	MCQ	+1	**A**
50	MCQ	+1	**B**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for incorrect attempts. Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.